

claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde
xer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-
a595885fdc3994f31.jsonl`
- SHA-256: `9f08e57fccd8ce3f5e7d6ef48ed312f1364da3a20935980fcee92ee26acecec3`
- Source modified: `2026-06-19T15:53:43+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`
```

Transcript

1. user (2026-06-19T15:52:06.166Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.

HARD RULES:

- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.

Working dir is the repo root; use Read/Grep/Glob freely.

ADVERSARIALLY VERIFY these proposed staleness fixes for ONE file:

docs/RALLY_DETECTION_GAP_CLOSURE.md.

For EACH proposed fix: (1) open docs/RALLY_DETECTION_GAP_CLOSURE.md and confirm stale_text appears VERBATIM (tighten/expand it to an exact, UNIQUE match if needed ? this text will be applied by exact string match, so it must be perfect); (2) re-open the cited code yourself and confirm the claim is genuinely contradicted by the CURRENT code; (3) confirm corrected_text is accurate, preserves voice, and is minimal; (4) DROP it if it is subjective, an owner-gated decision, already-correct, or you cannot verify the evidence. Output ONLY the survivors (verdict confirm or revise), with stale_text guaranteed to be an exact unique substring of the file.

PROPOSED FIXES:

```
[
  {
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": " - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle
box ? person box = held) `[design]` from the fusion plan §4,",
    "corrected_text": " - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation`
(shuttle box ? person box = held) `[verified]` built in `fusion_features.py` (default-OFF,
unmeasured on golden) from the fusion plan §4,",
    "why_stale": "Tags shuttle_player_colocation `[design]` (proposed), but the feature is built
in backend/pipeline/detectors/fusion_features.py:168 (in PERSON_FEATURE_NAMES, computed by
compute_person_features) and wired in fusion_hybrid.py under cue_flags.person. Per this doc's
own honesty key, `[design]` is only correct for proposed-not-built; the feature is built but
default-OFF and unmeasured on golden (the generalized rally-END rule that consumes it is what
remains design).",
    "severity": "high"
  },
  {
```

```

    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "person producer in `yolo_hybrid` (built) . `shuttle_player_colocation` (new
feature) . `scratch/ingest_golden.py`",
    "corrected_text": "person producer in `yolo_hybrid` (built) . `shuttle_player_colocation`
(built in `fusion_features.py`, default-OFF + unmeasured on golden) .
`scratch/ingest_golden.py`",
    "why_stale": "The reuse map labels shuttle_player_colocation \"(new feature)\" in a \"what's
new vs there\" list, implying not built. It is built in
backend/pipeline/detectors/fusion_features.py:168 (PERSON_FEATURE_NAMES line 197) and persisted
by fusion_hybrid.py::_person_features. What is genuinely undone is extracting/measuring it on
the golden clips (this doc's §7 row P3), not building the feature.",
    "severity": "medium"
},
{
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "***Caveat:** two golden clips so far (n=13 corpus, GX010141+GX010137) ?
directional, not conclusive.",
    "corrected_text": "***Caveat:** two golden clips so far (n=15 corpus, GX010141+GX010137) ?
directional, not conclusive.",
    "why_stale": "The parenthetical states the current corpus size as 'n=13 corpus', but
output/human_lovo_manifest.json now has 15 entries. The 'two golden clips so far' (the n=2
ground-truth-scored finding) is left intact; only the current-corpus-size token is corrected.",
    "severity": "medium"
},
{
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "- **n is small.** One golden clip scored so far (GX010141); the corpus is
n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more
(GX010137 / GX020140 / a large-count clip next).",
    "corrected_text": "- **n is small.** One golden clip scored so far (GX010141); the corpus is
n=15 with several weak clips. Per-video conclusions firm up only as the owner labels more
(GX010137 / GX020140 / a large-count clip next).",
    "why_stale": "Present-tense statement 'the corpus is n=12', but
output/human_lovo_manifest.json now has 15 entries. Only the current-corpus-size token is
corrected; the 'one golden clip scored so far' progress wording is left to the doc's own §3
which already supersedes it.",
    "severity": "medium"
},
{
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12
corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |",
    "corrected_text": "| D3 | **Golden labels = ground truth; Gemini = reference, not truth** |
n=15 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |",
    "why_stale": "The decision-source column characterizes the current corpus as 'n=12 corpus',
but output/human_lovo_manifest.json now has 15 entries. The decision itself is unchanged; only
the stale corpus-size figure is corrected.",
    "severity": "low"
}
]

```

2. assistant (2026-06-19T15:52:08.633Z)

I'll verify each proposed fix by confirming the stale_text appears verbatim in the doc, then checking the cited code.

3. user (2026-06-19T15:52:09.599Z)

```

1      # Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)
2
3      > **Status:** `DRAFT` ? implementation **owner-gated** . **Owner:** Avi . **Created:**
2026-06-18 . **Last updated:** 2026-06-18
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory

```

```

`current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
6      > **Relation to existing docs:** **peer-of**
[ `RALLY_QUALITY_RESEARCH.md` ](RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework
doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on**
[ `MULTI_SIGNAL_FUSION_PLAN.md` ](MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics),
[ `QUALITY_ITERATIONS.md` ](QUALITY_ITERATIONS.md) (the R-log / empirical backbone),
[ `GOLDEN_VIDEOS.md` ](GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) /
`[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read
$6.
8
9      ---
10
11     ## 0. TL;DR
12     We are using the **growing golden corpus** (human rally labels) to find and close the
rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip
already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a
real rally, its edges are tight ( $|?| < 1$  s). **The remaining gap is DETECTION, not boundaries:**
(G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision**
? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-
extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity
alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the
priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall**
(opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1
lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the
player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style
precision) as the durable win. **Caveat:** two golden clips so far (n=13 corpus,
GX010141+GX010137) ? directional, not conclusive.
13
14     ## 1. Problem & goal
15     The detector is good enough to ship a highlights product, but the golden eval shows
clip-dependent accuracy that we want to **close consistently on every video, not just on
average** ? catching the failure modes in the quality loop instead of after launch.
16
17     - **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs
the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set
discipline, [[eval-dual-set-regression]]).
18     - **"Good" =** the three gaps below closed enough that a milestone flip (default-ON)
clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report
([[launch-report-convention]]).
19     - **Read first to get current:**
[ `RALLY_QUALITY_RESEARCH.md` ](RALLY_QUALITY_RESEARCH.md) $1?2 (the gap + diagnosis),
[ `MULTI_SIGNAL_FUSION_PLAN.md` ](MULTI_SIGNAL_FUSION_PLAN.md) $4 (in-flight vs in-hand),
[ `HOW_RALLY_DETECTION_WORKS.md` ](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).
20
21     ## 2. Decisions locked
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | The gap is **DETECTION (recall + precision + rally-END), not boundary precision**
| golden GX010141, 2026-06-18 `[verified]` | Don't build a generic boundary head; spend on
rally-state signals. |
25     | D2 | Measure **per-video AND aggregate; no regression on ANY video** | [[eval-dual-
set-regression]] / [[decision-rigor-norm]] | A win must broadly help, not trade one clip for
another. |
26     | D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus |
Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |
27     | D4 | New signals land **default-OFF + flag-gated**, promoted only on measured lift |
[ `EXPERIMENT_HARNESS.md` ](EXPERIMENT_HARNESS.md) / [[weak-signals-retained]] | Zero-regression
rollout; retained even on a small-n null. |
28     | D5 | **Rally-END needs a PLAYER-state signal** ? shuttle velocity can't separate a
competitive rally from a casual post-point return | owner insight + R4 limitation, 2026-06-18
`[design]` | Add person kinematics / shuttle-in-hand as the rally-end discriminator. |
29
30     ## 3. Foundation ? the measured gaps (golden ground truth)
31     Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone

```

windowing = cap6 + R4 ([`QUALITY\_ITERATIONS.md`])(QUALITY_ITERATIONS.md) R3/R4). Local & Gemini vs truth, n=2 so far: `[verified]`

32

33 | clip (golden) | system | found | tIoU@0.5 P/R/F1 | tolF1 | hits | false+ | missed |
 \?start\ | \?end\ |

34 |---|---|---|---|---|---|---|---|

35 | **GX010141** (29, hard/drilling) | LOCAL | 22 | 0.59/0.45/**0.51** | 0.43 | 13 | 11 |
 16 | 0.78 | 0.68 |

36 | | GEMINI | 16 | 0.94/0.52/**0.67** | 0.53 | 15 | 4 | 14 | 0.31 | 0.46 |

37 | **GX010137** (34, normal match) | LOCAL | 39 | 0.77/**0.88**/**0.82** | 0.68 | 30 | 14
 | 9 | 0.39 | 0.44 |

38 | | GEMINI | 27 | 1.00/0.79/**0.89** | 0.89 | 27 | 0 | 7 | 0.50 | 0.52 |

39

40 - **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25
 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above
 the mean. `[verified]`

41 - **Opposite failure modes ? the central finding (n=2).** **LOCAL** = high recall, low
 precision; **GEMINI** = high precision, low recall. **On the normal clip GX010137, local's recall**
0.88 BEATS Gemini's 0.79 (local finds more real rallies) but local fires 14 false positives;**
Gemini fires zero false positives but misses 7 rallies. They are complementary (on
 GX010141 they overlap on only ~3 rallies yet each hits ~13/15 of the golden set). **? fusion**
(local's recall + Gemini-style precision filtering) is the clear win.

42 - **The gaps, re-prioritized by the data:**

43 - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39
 false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter
 local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-
 recalls it.

44 - **G1 ? recall** is clip-dependent: a local **strength** on normal play (? Gemini), but
 on hard continuous-drilling clips **both** miss ~half (GX010141 recall ~45/52%). The recall lever
 matters mainly for the hard clips.

45 - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ?
 window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false
 positives/merges.

46 - **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ?
 normal clip ??0.07 (0.82 vs 0.89), and local **out-recalls** Gemini there. The deficit is
 precision, not detection.

47 - **What is NOT the gap:** boundary precision. When local hits a rally its edges are
 tight (|?start|/?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing
 is doing its job. `[verified]`

48 - **Long-rally lens refinement (P7, n=15, [verified]):** filtering the eval to >5s
 rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-**
dominated (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt** still over-fires 1.58x
 on LONG rallies (adjacent-court games are genuine long rallies). ? the multicourt precision
 gap needs **court-localization** (Gate-B / court mask, Lever C), NOT a duration filter; and the
 strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not
 just short FPs. See [`QUALITY\_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".

49 - ? **Data caveat:** GX010141 was labeled before ~3 rally-annotator-tool bugs were found
 + fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before
 drawing hard conclusions from it.

50

51 **## 4. Design / architecture ? the levers (REUSE MAP)**

52 The signal mechanics live in
 [`MULTI\_SIGNAL\_FUSION\_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which
 gap.

53

54 - **Lever A ? player-state rally-END rule** (targets **G3 + G2**) `[design]`. Generalize
 R4 from shuttle-only to **shuttle OR bilateral player motion**: end the rally at the last frame
 with competitive evidence from **either**, and **hard-terminate** on a "shuttle picked up" event.
 Two cheap detectors, no new model required:

55 - **Player kinematics** ? person velocity drops / players go stationary after a point
 (the casual returner taps it while the other walks). **Reuse:** the person cue producer
 (torchvision + ByteTrack) in `yolo\_hybrid` (fusion P1, `[verified]` exists).

56 - **Shuttle-in-hand / out-of-play** ? `shuttle\_player\_colocation` (shuttle box ? person
 box = held) `[design]` from the fusion plan S4, **or** the no-new-model proxy: **sustained WASB**
 `Visibility=0` after the last fast stroke (held shuttle goes dark). Use it as a **termination**

anchor + reverse-timeline search** to the true end (last net-crossing / last fast-shuttle / last bilateral motion) ? the owner's idea, refined: the hold event is *late*, so anchor-then-backtrack, don't use its timestamp as the end.

57 - **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ? detector sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-fusion** angle below. Measure recall floor per video; the highest-leverage labels are the *missed* rallies (active learning, RALLY_QUALITY_RESEARCH §4.3).

58 - **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built scaffolding: `rally\_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]` built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature path.

59 - **Pure-precision cleanup** (**G2**): `rally\_gate` Gate A kills single-feed / held-shuttle *starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and independent of the person stack.

60

61 **Reuse map (what's new vs there):** `rally\_gate.py::gate\_windows` (Gate A, built) · `tracknet\_runner` R4 end-trim (extend, don't replace) · `backend/eval/windowing.py` (preset knobs) · person producer in `yolo\_hybrid` (built) · `shuttle\_player\_colocation` (new feature) · `scratch/ingest\_golden.py` + `scratch/boxhill17\_score\_golden.py` (the per-video golden scoring loop, `[verified]`).

62

63 ## 6. Honest limits ? what this does NOT do

64 - **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).

65 - **Nothing here is shipped or even started** ? implementation is owner-gated. A DONE row in §7 means *measured on golden*, not *flipped on by default*.

66 - **Player-state levers need GPU person-detection on the golden clips** (not yet extracted); until then Lever A/B are design, not data.

67 - The boundary tightness is from clips where local *hits* the rally; it says nothing about the rallies it misses (G1).

68

69 ## 7. Deliverables & progress tracker ? **source of truth**

70 Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.

71

ID	Deliverable	Depends on	Gated?	Status	PR
P0	Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141 ingested (#12) + scored	? No	? this work		
P1	Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ? per-video baseline table owner labels	? owner	? ?		
P2	Wire `rally_gate` **Gate A** (net-crossings) to the served path ? cheapest G2 win	? ? owner	? ?		
P3	Extract person-velocity + `shuttle_player_colocation` on the golden clips (default-OFF features)	P1	? owner	? ?	
P4	**Lever A**: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate / ?\ ?end\ vs R4	P3	? owner	? ?	
P5	**Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression	P3	? owner	? ?	
P6	Launch decision: milestone + best levers default-ON, with a Launch Report (both rulers + over-seg guard)	P4,P5	? owner	? ?	
P7	**Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See `[QUALITY_ITERATIONS.md]` (QUALITY_ITERATIONS.md) "P7".	? No (eval+cfg, default-OFF)	? measured	this PR	

82

83 ## 8. Open questions ? owner / external

84 1. **Which lever first** once labels land ? the no-model wins (Gate A wiring + shuttle-invisibility end-anchor), or go straight to person-velocity? *(decides: owner, after the per-video baseline)*

85 2. **Per-video target** ? "at-par-with-Gemini-vs-golden on every clip," or an absolute

```

F1 floor? *(decides: owner)*
86      3. **Recall lever** ? detector-sensitivity tuning vs ensemble fusion as the primary path
for G1. *(decides: data, post-P3)*
87
88      ## 9. Definition of done
89      - [ ] Golden corpus covers the Boxhill set + enough variety that per-video reads are
trustworthy (n meaningfully > 12).
90      - [ ] The **rally-END quick-return** failure mode (G3) is closed: measured ? merge-rate
+ ? |?end| on the clips that exhibit it, no regression elsewhere.
91      - [ ] Local is **at-par-or-better than Gemini vs golden on every corpus video** (no per-
video F1 regression).
92      - [ ] The winning levers are flipped **default-ON** via a Launch Report (over-seg guard
clean on both rulers).
93      - [ ] ? flip `Status ? DONE`, archive to `docs/archives/`.
94
95      ## 10. References
96      **Code anchors `[verified]`:** `backend/pipeline/detectors/rally_gate.py` (Gate A) ·
`backend/pipeline/detectors/tracknet_runner.py` (R4 end-trim, `_trim_inactive_tail`) ·
`backend/eval/windowing.py` · `backend/eval/rally_seg_eval.py` + `tolerance_metrics.py` (R2) ·
`backend/pipeline/segmenters/yolo_hybrid.py` (person cue) · `scratch/ingest_golden.py`,
`scratch/boxhill17_score_golden.py`.
97      **Internal docs:** `[RALLY_QUALITY_RESEARCH.md]` (RALLY_QUALITY_RESEARCH.md),
`[MULTI_SIGNAL_FUSION_PLAN.md]` (MULTI_SIGNAL_FUSION_PLAN.md) §4,
`[QUALITY_ITERATIONS.md]` (QUALITY_ITERATIONS.md),
`[EXPERIMENT_HARNESS.md]` (EXPERIMENT_HARNESS.md), `[GOLDEN_VIDEOS.md]` (GOLDEN_VIDEOS.md),
`[HOW_RALLY_DETECTION_WORKS.md]` (HOW_RALLY_DETECTION_WORKS.md).
98
99      ---
100
101     ### Changelog
102     - 2026-06-18 ? Created. Seeded with the GX010141 golden scoring (first ground-truth
clip), the three named gaps (recall / precision / rally-END), and the player-state end-rule
design from the owner's quick-return insight.
103     - 2026-06-18 ? Added GX010137 (golden #13, normal match play): local F1@0.5 **0.82**
(recall **0.88 > Gemini 0.79**) vs Gemini 0.89. Confirms **opposite failure modes** (local high-
recall/low-precision, Gemini the reverse) ? **precision is local's #1 lever**, fusion the
durable win. Flagged the GX010141 pre-annotator-fix data caveat.
104     - 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF),
measured n=15.** Eval `--min-duration/--stratify` (`rally_seg_eval.py`) + product
`stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio
1.43±1.04) but **multicourt over-fires 1.58× even on long rallies** (? needs court-localization,
Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 +
added §7 row P7.
105

```

4. user (2026-06-19T15:52:12.577Z)

fatal: Cannot fast-forward to multiple branches.

5. assistant (2026-06-19T15:52:15.461Z)

The doc is loaded. Note the repo has uncommitted/local state and a pull issue, but I'll proceed read-only on what's local. Now let me verify the cited code for each claim.

6. user (2026-06-19T15:52:15.967Z)

```

1      """Pure per-window person-cue features for multi-signal rally fusion
(MULTI_SIGNAL_FUSION_PLAN §3.2/§4).
2
3      NO I/O, no torch, no cv2 ? mirrors `rally_gate.py`'s discipline so the whole feature
4      math is unit-testable in a GPU-free dev container. The detection (person boxes,
5      shuttle trajectory) is produced elsewhere (`PersonDetector`, the WASB runner); this
6      module only turns those signals into a handful of **scale/translation-invariant**
7      features (counts / ratios / zone-membership ? never raw pixel coordinates, which would

```

```

8     memorise this court/camera and fail to generalise off a small labelled set).
9
10    Conventions:
11    - A person box is ``(x1, y1, x2, y2)`` in **pixels**; ``(track_id, box)`` when motion
needs it.
12    - ``per_frame`` is ``{frame_idx: [(track_id, box), ...]}`` keyed by the ORIGINAL video
frame
13        index (so it aligns directly with the shuttle trajectory's ``.frame`` ? no clip re-
timing).
14    - A shuttle point has ``.frame``, ``.visible``, ``.x``, ``.y`` (pixels) ? the WASB
trajectory shape.
15    - ``court_polygon`` is normalised 0-1 ``[(x, y), ...]`` or ``None`` (None ? no mask,
every
16        detection counts ? the player-count-only mode).
17    - ``net`` is ``(axis 'x'|'y', position 0-1 normalised)`` or ``None`` (None ? two-sided =
0.0).
18
19    Every function degrades gracefully (returns 0.0 / empty) on missing inputs; outputs are
20    floats in [0, 1] except ``mean_player_motion`` (a non-negative normalised speed).
21    """
22    from __future__ import annotations
23
24    from typing import Dict, List, Optional, Sequence, Tuple
25
26    BBox = Tuple[float, float, float, float]
27    Net = Tuple[str, float]
28
29
30    def point_in_polygon(point: Tuple[float, float], polygon: Sequence[Tuple[float, float]])
-> bool:
31        """Ray-casting point-in-polygon. `point` and `polygon` share a coordinate space
32        (we use normalised 0-1). Empty/degenerate polygon (<3 vertices) ? False."""
33        if polygon is None or len(polygon) < 3:
34            return False
35        x, y = point
36        inside = False
37        n = len(polygon)
38        j = n - 1
39        for i in range(n):
40            xi, yi = polygon[i]
41            xj, yj = polygon[j]
42            # Does the horizontal ray at y cross edge (i, j)?
43            if (yi > y) != (yj > y):
44                x_cross = (xj - xi) * (y - yi) / (yj - yi) + xi if yj != yi else xi
45                if x < x_cross:
46                    inside = not inside
47            j = i
48        return inside
49
50
51    def _foot_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float,
float]:
52        """Normalised foot point (bottom-centre) of a person box ? where the player stands,
53        the right anchor for court membership. Returns (0,0) if frame dims are unusable."""
54        if frame_width <= 0 or frame_height <= 0:
55            return (0.0, 0.0)
56        x1, _y1, x2, y2 = box
57        return ((x1 + x2) / 2.0 / frame_width, y2 / frame_height)
58
59
60    def _center_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float,
float]:
61        if frame_width <= 0 or frame_height <= 0:
62            return (0.0, 0.0)
63        x1, y1, x2, y2 = box

```

```

64         return ((x1 + x2) / 2.0 / frame_width, (y1 + y2) / 2.0 / frame_height)
65
66
67     def person_in_court(box: BBox, frame_width: float, frame_height: float,
68                         court_polygon: Optional[Sequence[Tuple[float, float]]]) -> bool:
69         """Is this person on the court? Foot-point-in-polygon when a polygon is supplied;
70         True (count everyone) when it is None. Unusable frame dims (w/h <= 0) make the foot
71         point meaningless, so a masked check returns False rather than testing a bogus
72         (0,0)."""
73         if court_polygon is None:
74             return True
75         if frame_width <= 0 or frame_height <= 0:
76             return False
77         return point_in_polygon(_foot_norm(box, frame_width, frame_height), court_polygon)
78
79     def in_court_counts(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
80                        frame_height: float,
81                        court_polygon: Optional[Sequence[Tuple[float, float]]]) ->
82     List[int]:
83         """Per-frame count of in-court persons (one entry per sampled frame)."""
84         return [
85             sum(1 for _tid, box in dets if person_in_court(box, frame_width, frame_height,
86                 court_polygon))
87             for _frame_idx, dets in sorted(per_frame.items())
88         ]
89
90     def _median(vals: Sequence[float]) -> float:
91         if not vals:
92             return 0.0
93         s = sorted(vals)
94         n = len(s)
95         return float(s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2]))
96
97     def players_in_court(counts: Sequence[int]) -> float:
98         """Median per-frame in-court count (?2 singles / 4 doubles; 0-1 = dead time)."""
99         return _median(list(counts))
100
101
102     def in_court_ratio(counts: Sequence[int], min_players: int = 2) -> float:
103         """Fraction of sampled frames with >= `min_players` in court (track stability vs
104         flicker)."""
105         if not counts:
106             return 0.0
107         return sum(1 for c in counts if c >= min_players) / len(counts)
108
109     def mean_player_motion(per_frame: Dict[int, List[Tuple[int, BBox]]],
110                           frame_width: float, frame_height: float) -> float:
111         """Mean per-track centre displacement between consecutive sampled frames, with x
112         normalised by frame width and y by frame height (per-axis; uniform-scale-invariant,
113         not aspect-ratio-isotropic). 0.0 if <2 frames or no shared tracks."""
114         if frame_width <= 0 or len(per_frame) < 2:
115             return 0.0
116         frames = sorted(per_frame.keys())
117         steps: List[float] = []
118         for a, b in zip(frames, frames[1:]):
119             prev = {tid: box for tid, box in per_frame[a]}
120             for tid, box in per_frame[b]:
121                 if tid in prev:
122                     cx0, cy0 = _center_norm(prev[tid], frame_width, frame_height)
123                     cx1, cy1 = _center_norm(box, frame_width, frame_height)
124                     steps.append(((cx1 - cx0) ** 2 + (cy1 - cy0) ** 2) ** 0.5)

```



```

125         if not steps:
126             return 0.0
127         return sum(steps) / len(steps)
128
129
130     def two_sided_motion(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
131                          frame_height: float, net: Optional[Net],
132                          court_polygon: Optional[Sequence[Tuple[float, float]]]) -> float:
133         """Fraction of sampled frames with >=1 in-court person on BOTH net halves.
134
135         A real rally is two-sided; a single player feeding/holding the shuttle is one-sided.
136         The net splits the play axis at `net=(axis, position)` in normalised coords. None ?
137         0.0.
138         """
139         if net is None or not per_frame:
140             return 0.0
141         axis, pos = net
142         both = 0
143         for _frame_idx, dets in per_frame.items():
144             near, far = False, False
145             for _tid, box in dets:
146                 if not person_in_court(box, frame_width, frame_height, court_polygon):
147                     continue
148                 fx, fy = _foot_norm(box, frame_width, frame_height)
149                 coord = fx if axis == "x" else fy
150                 if coord < pos:
151                     near = True
152                 else:
153                     far = True
154             if near and far:
155                 both += 1
156         return both / len(per_frame)
157
158     def _shuttle_in_box(sx: float, sy: float, box: BBox, frame_width: float, frame_height:
159 float,
160                       pad_frac: float) -> bool:
161         """Is the (pixel) shuttle point inside a person box, expanded by `pad_frac` of frame
162         width/height (the 'adjacent / in-hand' tolerance)?"""
163         x1, y1, x2, y2 = box
164         px = pad_frac * frame_width
165         py = pad_frac * frame_height
166         return (x1 - px) <= sx <= (x2 + px) and (y1 - py) <= sy <= (y2 + py)
167
168     def shuttle_player_colocation(per_frame: Dict[int, List[Tuple[int, BBox]]],
169 shuttle_points,
170                                   frame_width: float, frame_height: float,
171                                   pad_frac: float = 0.03) -> float:
172         """Fraction of (visible shuttle frames that were also person-sampled) in which the
173         shuttle sits inside/adjacent to a person box. **held/in-hand ? high; in-flight ?
174         ~0**
175
176         ? the direct held-shuttle discriminator. 0.0 if no aligned visible shuttle frames.
177         """
178         if not per_frame or frame_width <= 0 or frame_height <= 0:
179             return 0.0
180         aligned = 0
181         colocated = 0
182         for p in shuttle_points or []:
183             if not getattr(p, "visible", False):
184                 continue
185             dets = per_frame.get(int(p.frame))
186             if dets is None:
187                 # shuttle frame wasn't person-sampled ? can't judge
188                 continue
189             aligned += 1

```

```

186         if any(_shuttle_in_box(p.x, p.y, box, frame_width, frame_height, pad_frac)
187                 for _tid, box in dets):
188             colocated += 1
189     return colocated / aligned if aligned else 0.0
190
191
192     # The ~5 person features persisted into candidate `stats` (order = the learned-fusion
193     # feature columns appended after the shuttle features in distill_local.FEATURE_NAMES).
194     PERSON_FEATURE_NAMES = [
195         "players_in_court", "two_sided_motion", "mean_player_motion",
196         "in_court_ratio", "shuttle_player_colocation",
197     ]
198
199
200     def compute_person_features(per_frame: Dict[int, List[Tuple[int, BBox]]],
201                                shuttle_points,
202                                frame_width: float, frame_height: float,
203                                court_polygon: Optional[Sequence[Tuple[float, float]]] =
204                                None,
205                                net: Optional[Net] = None,
206                                min_players: int = 2) -> Dict[str, float]:
207         """Aggregate the per-window person features from per-frame detections + the in-
208         window
209         shuttle trajectory. Pure; returns exactly `PERSON_FEATURE_NAMES`, all floats."""
210         counts = in_court_counts(per_frame, frame_width, frame_height, court_polygon)
211         return {
212             "players_in_court": players_in_court(counts),
213             "two_sided_motion": two_sided_motion(per_frame, frame_width, frame_height, net,
214             court_polygon),
215             "mean_player_motion": mean_player_motion(per_frame, frame_width, frame_height),
216             "in_court_ratio": in_court_ratio(counts, min_players),
217             "shuttle_player_colocation": shuttle_player_colocation(
218                 per_frame, shuttle_points, frame_width, frame_height),
219         }

```

7. user (2026-06-19T15:52:17.327Z)

EXISTS

8. assistant (2026-06-19T15:52:24.332Z)

`shuttle\_player\_colocation` is indeed BUILT in `fusion\_features.py` (function at line 168, in `PERSON\_FEATURE\_NAMES` line 197, computed by `compute\_person\_features` line 213). Now let me verify it's wired in `fusion\_hybrid.py` and check the manifest count.

9. user (2026-06-19T15:52:24.932Z)

```

list
len: 15
[{'name': 'adarsh_avi_singles', 'trajectory': 'C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv', 'fps': 59.94, 'frame_width': 1920,
'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv'}, {'name': 'kushagra_singles', 'trajectory':
'C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/kushagra_traj.csv', 'fps': 59.94,
'frame_width': 1920, 'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv'}, {'name': 'largetest_doubles', 'trajectory':
'C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/largetestvideo_traj.csv', 'fps':
29.97, 'frame_width': 1920, 'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv'}]

```

10. user (2026-06-19T15:52:25.124Z)

backend\pipeline\segmenters\fusion_hybrid.py

11. assistant (2026-06-19T15:52:27.382Z)

The manifest has 15 entries. Now let me verify the `fusion\_hybrid.py` wiring referenced in the proposed fixes (`cue\_flags.person`, `\_person\_features`).

12. user (2026-06-19T15:52:29.068Z)

```
10: is explicitly enabled (`indexing.fusion.cue_flags.person`), the person-cue features
11: (`fusion_features`), persisting them into `candidate_segments.stats` for the learned
36: from backend.pipeline.detectors import fusion_features as ff
88:         if fcfg.get("cue_flags", {}).get("person", False):
89:             stats.update(self._person_features(cw, points, fps, frame_width, video_path,
fcfg))
93:     # Person features (from _person_features when cue_flags.person) are now persisted for the
harness
111:     def _person_features(self, cw: Dict[str, Any], points: List[Any], fps: float,
132:         return {k: 0.0 for k in ff.PERSON_FEATURE_NAMES}
138:         return ff.compute_person_features(det["frames"], shuttle_in, fw, fh,
160:         person_on = bool(fcfg.get("cue_flags", {}).get("person", False))
```

13. assistant (2026-06-19T15:52:43.795Z)

The wiring is confirmed: `fusion\_hybrid.py:88` gates on `cue\_flags.person`, calls `\_person\_features` (line 89/111), which calls `ff.compute\_person\_features` (line 138), which produces `shuttle\_player\_colocation`. It's built but default-OFF (gated on a flag that defaults to `False`).

Now I have all the evidence I need. Let me verify each proposed fix's stale_text is a verbatim unique match and assess corrections.

Fix 1 ? stale_text: ` - \*Shuttle-in-hand / out-of-play\* ? `shuttle_player_colocation` (shuttle box ? person box = held) `[design]` from the fusion plan §4, ? This appears verbatim on line 56. The `[design]` tag IS stale: the feature is built (`fusion_features.py:168`). Per the doc's honesty key (line 7), `[design]` = "proposed". The corrected\_text adds `[verified]` + accurate built/default-OFF/unmeasured qualifier. This is accurate but I'll tighten ? the corrected text says "built in `fusion_features.py`" which matches reality. Good. CONFIRM.

Fix 2 ? stale_text: `person producer in `yolo_hybrid` (built) . `shuttle_player_colocation` (new feature) . `scratch/ingest_golden.py` ? appears verbatim on line 61. "(new feature)" in a "what's new vs there" list implies not-yet-built, but it IS built. The correction is accurate. CONFIRM.

Fix 3 ? stale_text: `\*\*Caveat:\*\* two golden clips so far (n=13 corpus, GX010141+GX010137) ? directional, not conclusive.` ? appears verbatim on line 12. Manifest has 15 entries. n=13 ? n=15. The "two golden clips so far" (n=2 ground-truth-scored) is preserved. CONFIRM.

Fix 4 ? stale_text references line 64: `the corpus is n=12`. Let me confirm the full verbatim string.

The line 64 reads: `- \*\*n is small.\*\* One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).` ? matches the proposed stale_text verbatim. n=12 ? n=15 is correct per manifest. CONFIRM.

Fix 5 ? stale_text line 26: `| D3 | \*\*Golden labels = ground truth; Gemini = reference, not truth\*\* | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |` ? matches verbatim. n=12 ? n=15. CONFIRM.

Let me double-check there's consistency on the corpus-size figures across the doc ? there are several `n=12`/`n=13`/`n=15` references, and I want to ensure the manifest is the right ground-truth source for "corpus" size and that my matches are unique.

14. user (2026-06-19T15:52:45.441Z)

```

12:[Omitted long matching line]
26:| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus | Gemini
under-detects too ? agreement ? accuracy. Report both, gate on golden. |
40:- **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25
0.633; R2 toIF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above
the mean. `[verified]`
48:[Omitted long matching line]
64:- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several
weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a
large-count clip next).
81:| P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob
(`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is
short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58x even on long
rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-
contaminated). See [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg,
default-OFF) | ? measured | this PR |
104:- 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF), measured
n=15.** Eval `--min-duration`/`--stratify` (`rally_seg_eval.py`) + product
`stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio
1.43?1.04) but **multicourt over-fires 1.58x even on long rallies** (? needs court-localization,
Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 +
added §7 row P7.

```

15. assistant (2026-06-19T15:52:53.580Z)

I need to be careful here. There's a subtlety with the corpus-size figures. The doc has multiple distinct numbers:

- Line 12, 26, 64: `n=12`/`n=13` referring to "corpus"
- Line 40: `Corpus aggregate (n=13, milestone)` ? a measured aggregate scored over a specific corpus
- Lines 81, 104: `n=15` for the P7 long-rally measurement

The proposed fixes update `n=12`/`n=13` ? `n=15` based on the manifest having 15 entries. But I must verify the manifest is genuinely "the corpus" referenced. The P7 measurement (line 81) already uses `n=15` and was the most recent work. Let me confirm the manifest is what these "corpus" figures track, and check whether line 40's `n=13` (the measured aggregate `0.474 [0.359, 0.579]`) is a different, frozen measurement that should NOT be bumped.

The proposed fixes deliberately do NOT touch line 40 (the measured aggregate with CI) ? that's a specific measurement frozen at its n. Good. But I need to confirm that the "corpus" the prose refers to (lines 12, 26, 64) is indeed the manifest, and that 15 is the right current number. Let me check the manifest names and the P7/QUALITY_ITERATIONS provenance for "n=15".

16. user (2026-06-19T15:52:55.289Z)

```

adarsh_avi_singles
kushagra_singles
largetest_doubles
gbaaddy
testlarge_short
mahadevpura_singles
mahadevpura_2
GX010128
mahadevpura_1
GX030094
Badminton_BXH_2
GX010141
GX010137
Boxhill_Doubles
GX020094

```

17. user (2026-06-19T15:52:56.543Z)

---STATUS---

18. assistant (2026-06-19T15:53:11.161Z)

The manifest is committed (not dirty), has 15 entries, and includes both GX010141 and GX010137. The P7 work (line 81) measured `n=15` against this same corpus, confirming 15 is the current corpus size.

Now let me cross-check that the doc's own prose treats "the corpus" as this LOVO manifest. The P7 row (line 81) and changelog (line 104) both cite `n=15` as the current measured corpus, while lines 12/26/64 say `n=12`/`n=13` ? an internal inconsistency within the same document that confirms the lower numbers are stale relative to the latest P7 measurement. This is strong evidence.

Let me verify one more time that line 40's `n=13` aggregate is genuinely a frozen measurement that the fixes correctly leave alone (it carries a specific CI `[0.359, 0.579]`), versus the prose "corpus" figures. The proposed fixes do not touch line 40 ? consistent with it being a frozen measured value. I'll note this is the correct scoping.

All five fixes verified. Let me confirm each stale_text is a unique substring (no duplicates).

- Fix 1 (line 56): the `\*Shuttle-in-hand / out-of-play\*` phrase appears only once. Unique.
- Fix 2 (line 61): `person producer in \yolo\_hybrid` (built) appears only once. Unique.
- Fix 3 (line 12): `\*\*Caveat:\*\* two golden clips so far (n=13 corpus` appears only once. Unique.
- Fix 4 (line 64): `\*\*n is small.\*\* One golden clip scored so far` appears only once. Unique.
- Fix 5 (line 26): the D3 row appears only once. Unique.

All confirmed. For the corpus-size fixes (3, 4, 5), I'll mark them `revise` only if I'd change the correction; the proposed corrections are accurate and minimal, so they are `confirm`. For fixes 1 and 2, the `[design]?built` corrections are accurate but I'll tighten fix 1's corrected_text slightly to preserve the doc's exact tag style ? the proposed correction is faithful, so `confirm`.

Let me do a final adversarial check on whether the manifest count is truly "the corpus" or whether some entries are non-golden (e.g., the corpus aggregate at line 40 used n=13). The fact that two entries ? `GX010137` and `GX010141` ? are the ground-truth-scored ones (n=2), while the full LOVO manifest is the broader corpus, means corpus=15 is now correct and the doc's older 12/13 figures are stale. This matches the doc's own P7 `n=15`.

All five fixes survive. Submitting.

19. user (2026-06-19T15:53:34.629Z)

Structured output provided successfully

20. assistant (2026-06-19T15:53:43.093Z)

All five proposed fixes survived adversarial verification.

Verified evidence:

- `shuttle\_player\_colocation` is genuinely BUILT ? `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\fusion\_features.py:168` (function), `:197` (`PERSON\_FEATURE\_NAMES`), `:213` (`compute\_person\_features`), wired in `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\fusion\_hybrid.py:88` (gated on `cue\_flags.person`, default `False`) ? `:111` `\_person\_features` ? `:138` `compute\_person\_features`. So the `[design]?` (new feature) tags (fixes 1, 2) are stale; corrected to built-but-default-OFF + unmeasured-on-golden (no overclaim).
- Corpus size: `C:\Users\avidu\Projects\badminton-highlight-indexer\output\human\_lovo\_manifest.json` has 15 entries (committed, clean tree), and the doc's own most-recent P7 work (lines 81, 104) already uses `n=15` ? confirming the `n=12`/`n=13` prose figures (fixes 3, 4, 5) are stale. The frozen measured aggregate at line 40 (`n=13` with CI `[0.359, 0.579]`) was correctly left untouched.

All `stale\_text` strings are exact, unique substrings of `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\RALLY\_DETECTION\_GAP\_CLOSURE.md`. All five verdicts are `confirm`.